

6.3 Probability	understand the language of probability understand and use the probability scale understand and use estimates or measures of probability from theoretical models understand the concepts of a sample space and an event, and how the probability of an event happening can be determined from the sample space list all the outcomes for single events and for two successive events in a systematic way estimate probabilities from previously collected data calculate the probability of the complement of an event happening	Outcomes, equal likelihood, events, random $P(\text{certainty}) = 1$ $P(\text{impossibility}) = 0$ For the tossing of two coins, the sample space can be listed as: Heads (H), Tails (T) $(H, H), (H, T), (T, H), (T, T)$ $P(\text{not } A) = 1 - P(A)$
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	use the addition rule of probability for mutually exclusive events understand and use the term <i>expected frequency</i>.	$P(\text{Either } A \text{ or } B \text{ occurring}) = P(A) + P(B)$ when A and B are mutually exclusive Determine an estimate of the number of times an event with a probability of $\frac{2}{5}$ will happen over 300 tries
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Papers 3H and 4H (Higher Tier)

Content overview

Knowledge of the Foundation Tier content is assumed for students being prepared for the Higher Tier.

- Number
 - Numbers and the number system
- Algebra
 - Equations, formulae and identities
 - Sequences, functions and graphs
- Geometry
 - Shape, space and measure
 - Vectors and transformation geometry
- Statistics

Assessment overview

- Two written papers.
- Each paper is assessed through a two-hour examination set and marked by Edexcel.
- The total number of marks for each paper is 100.
- Each paper weighted at 50% of the qualification, targeted at grades A* – D.

Content

A01 Number and algebra		
1 Numbers and the number system		
	Students should be taught to:	Notes
1.1 Integers	See Foundation Tier.	
1.2 Fractions	See Foundation Tier.	
1.3 Decimals	convert recurring decimals into fractions	$0.\dot{3} = \frac{1}{3}$, $0.2333\ldots = \frac{21}{90}$
1.4 Powers and roots	understand the meaning of surds manipulate surds, including rationalising the denominator where the denominator is a pure surd use index laws to simplify and evaluate numerical expressions involving integer, fractional and negative powers evaluate Highest Common Factors (HCF) and Lowest Common Multiples (LCM)	Express in the form $a\sqrt{2}$: $\frac{2}{\sqrt{8}}$, $\sqrt{18} + 3\sqrt{2}$ Express in the form $a + b\sqrt{2}$: $(3 + 5\sqrt{2})^2$ Evaluate: $\sqrt[3]{8^2}$, $625^{-\frac{1}{2}}$, $(\frac{1}{25})^{\frac{3}{5}}$
1.5 Set language and notation	understand sets defined in algebraic terms understand and use subsets understand and use the complement of a set use Venn diagrams to represent sets and the number of elements in sets use the notation $n(A)$ for the number of elements in the set A use sets in practical situations	If A is a subset of B, then $A \subset B$ Use the notation A'
1.6 Percentages	use reverse percentages repeated percentage change solve compound interest problems	In a sale, prices were reduced by 30%. The sale price of an item was £17.50. Calculate the original price of the item. Calculate the total percentage increase when an increase of 30% is followed by a decrease of 20% To include depreciation

1.7 Ratio and proportion	See Foundation Tier.	
1.8 Degree of accuracy	solve problems using upper and lower bounds where values are given to a degree of accuracy	The dimensions of a rectangle are 12 cm and 8 cm to the nearest cm. Calculate, to 3 significant figures, the smallest possible area as a percentage of the largest possible area.
1.9 Standard form	express numbers in the form $a \times 10^n$ where n is an integer and $1 \leq a < 10$ solve problems involving standard form	$150\,000\,000 = 1.5 \times 10^8$
1.10 Applying number	See Foundation Tier.	
1.11 Electronic calculators	See Foundation Tier.	

2 Equations, formulae and identities		
	Students should be taught to:	Notes